

# Regular spacing of salt structures as a Rayleigh–Taylor signature: edge-corrected point-pattern and structure-factor analysis of the U.S. Gulf Coast and North German basins

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## Abstract

Salt structures rise by a buoyancy-driven (Rayleigh–Taylor) instability that, in theory, selects a characteristic wavelength and should therefore imprint a *regular* map-view spacing of diapirs and pillows. Empirically the record is mixed: several provinces show little or no characteristic wavelength, and Rayleigh–Taylor models often fail to reproduce observed spacing. Whether complete salt-structure *populations* are statistically regular in map view has rarely been tested with modern edge-corrected spatial statistics. We analyse two large open datasets — 1253 Gulf Coast salt diapirs (USGS) and 693 North German salt structures (BGR InSpEE) — using the Clark–Evans nearest-neighbour index, the pair correlation function  $g(r)$ , Ripley’s  $L$ , and the structure factor  $S(k)$  with number variance, against both complete-spatial-randomness (CSR) and inhomogeneous-Poisson null models. Pooled across a whole basin the structures are *clustered*, but this reflects the superposition of distinct sub-basins; within a single coherent loading-driven province the structures are significantly *regularly spaced* (Clark–Evans  $R = 1.10$ – $1.47$ ,  $p \approx 0.002$ ; robust to a density-gradient inhomogeneous-Poisson null), with hard-core exclusion and a 12–15 km characteristic spacing. Regularity strengthens with salt thickness (Glückstadt Graben > central basin > Gulf shelf), as expected for wavelength selection. The mature-diapir sub-class *alone* is clustered because mature diapirs concentrate in depocentres, so sub-class selection must be avoided. The structure factor shows genuine intermediate-range suppression ( $S \approx 0.42$  at a  $\sim 37$  km scale, below the gradient-matched null) but the long-wavelength limit is masked by the intrinsic basin density gradient, so strong hyperuniformity cannot be confirmed. The results provide a quantitative, statistically hardened map-view signature of Rayleigh–Taylor wavelength selection, together with explicit cautions on superposition, sub-class selection, and density gradients.

## 1 Introduction

Salt flows on geological timescales and, being less dense than its overburden, is gravitationally unstable: a buried salt layer overlain by denser sediment is subject to a Rayleigh–Taylor (RT) instability whose linear theory selects a dominant wavelength set primarily by the salt-layer thickness and the viscosity/strength contrast (Trusheim, 1960; Hudec & Jackson, 2007; Fernandez & Kaus, 2015). Because the fastest-growing mode has a finite wavelength, the classical expectation is that salt pillows and the diapirs that evolve from them should be *quasi-regularly spaced* in map view, each structure surrounded by a salt-withdrawal rim syncline that suppresses the growth of close neighbours.

Whether real provinces realise this prediction is, however, unsettled. Spacing/thickness relations are reproduced in some settings but not others; variogram and spectral analyses of certain provinces report essentially stochastic salt-surface topography with no uniform wavelength, and detailed studies find that simple RT models often fail to reproduce observed diapir spacing once sedimentation, basement structure and later shortening are accounted for (Jackson & Hudec, 2017). Real salt tectonics is further complicated by structural inheritance: where salt ascent is localised by pre-existing faults, structures tend to align and *cluster* rather than disperse. The map-view spacing of a complete salt-structure population is therefore an open empirical question, and one that is well posed for spatial point-pattern analysis.

Surprisingly, although nearest-neighbour and spacing statistics have long been used on individual salt provinces, complete salt-structure populations have rarely been examined with modern edge-corrected point-pattern statistics combined with the structure factor, explicit null models, and attention to the basin-scale density gradient that is intrinsic to salt provinces. Here we do so for two large, openly licensed datasets that together span the two end-member tectonic settings of passive-margin sediment-loading (U.S. Gulf Coast) and an intracratonic basin with both loading- and inversion-driven salt movement (North German Basin). We ask three questions: (i) are loading-driven salt-structure populations significantly regularly spaced, and is any regularity robust to the basin density gradient; (ii) does the strength of regularity scale with salt thickness, as RT wavelength selection predicts; and (iii) what is the *character* of the order — merely a liquid-like spacing, or a longer-range suppression of density fluctuations?

## 2 Data

**U.S. Gulf Coast (USGS).** We use the U.S. Geological Survey compilation *Salt Diapirs in the Gulf Coast* (gcdiaping; U.S. Geological Survey, 2021), itself derived from Martin (1980) and regional structural compilations. The dataset contains 1253 salt-diapir outline polygons (NAD83 geographic coordinates) spanning the onshore and offshore Gulf Coast from Texas to Florida and south to the Bay of Campeche. We take the area-weighted polygon centroid as the point representing each structure and project geographic coordinates to a local Cartesian frame (equiarectangular projection at the mean latitude); all distances are reported in kilometres.

**North German Basin (BGR InSpEE).** We use the BGR *InSpEE-Salzstrukturen* dataset (BGR, 2015; base mapping after Reinhold et al., 2008), an open shapefile (ETRS89 / UTM zone 32N, metres) of salt-structure outlines in Northern Germany. The 855 polygons resolve into 693 named structures (area-weighted centroid per structure). A companion internal-structure-type layer classifies each structure as a salt diapir (*Salzdiapir*,  $n = 159$ ), a compressively overprinted diapir ( $n = 107$ ), a salt pillow (*Salzkissen*,  $n = 205$ ), a double-salinar structure ( $n = 100$ ), a tectonic structure ( $n = 86$ ), or unassigned. We define the *halokinetic* population as the loading/ascent-driven structures (diapirs, compressively overprinted diapirs, pillows and double-salinar structures;  $n = 571$ ), excluding the purely tectonic and unassigned ones.

## 3 Methods

**Province subsetting.** A point pattern pooled over several distinct sub-basins is the superposition of independent processes and will generally appear clustered even if each sub-basin is internally

regular. We therefore report whole-basin statistics for context but base inference on coherent single-province sub-regions of approximately uniform tectonic setting.

**Nearest-neighbour index.** The Clark–Evans index (Clark & Evans, 1954) is  $R = \bar{d}_{\text{NN}}/\bar{d}_{\text{CSR}}$ , where  $\bar{d}_{\text{NN}}$  is the mean nearest-neighbour distance. To remove edge bias we estimate  $\bar{d}_{\text{CSR}}$  by Monte-Carlo simulation of complete spatial randomness (CSR) in the identical convex-hull window (here  $S = 499$  realisations), so that  $R > 1$  indicates dispersion/regularity and  $R < 1$  clustering relative to the matched null; a one-sided  $p$ -value for dispersion is obtained by rank.

**Second-order statistics.** The pair correlation function  $g(r)$  and Ripley’s  $K$  (reported as  $L(r) - r = \sqrt{K(r)/\pi} - r$ ) are computed with the border (reduced-sample) edge correction (Ripley, 1977; Baddeley et al., 2015).  $g(r) = 0$  at small  $r$  measures hard-core exclusion; the first  $g(r)$  maximum gives the characteristic spacing;  $L(r) - r < 0$  indicates regularity. Significance is assessed with the global studentised maximum-deviation envelope test (Myllymäki et al., 2017) over the interaction range, against (i) a CSR null and (ii) an inhomogeneous-Poisson null whose intensity is a Gaussian kernel estimate of the observed intensity (bandwidth  $2\times$  median nearest-neighbour distance). The inhomogeneous null builds the basin density gradient into the reference, so that a significant result is not an artifact of the gradient; because a smooth gradient biases the statistics toward *clustering*, detection of regularity against this null is conservative.

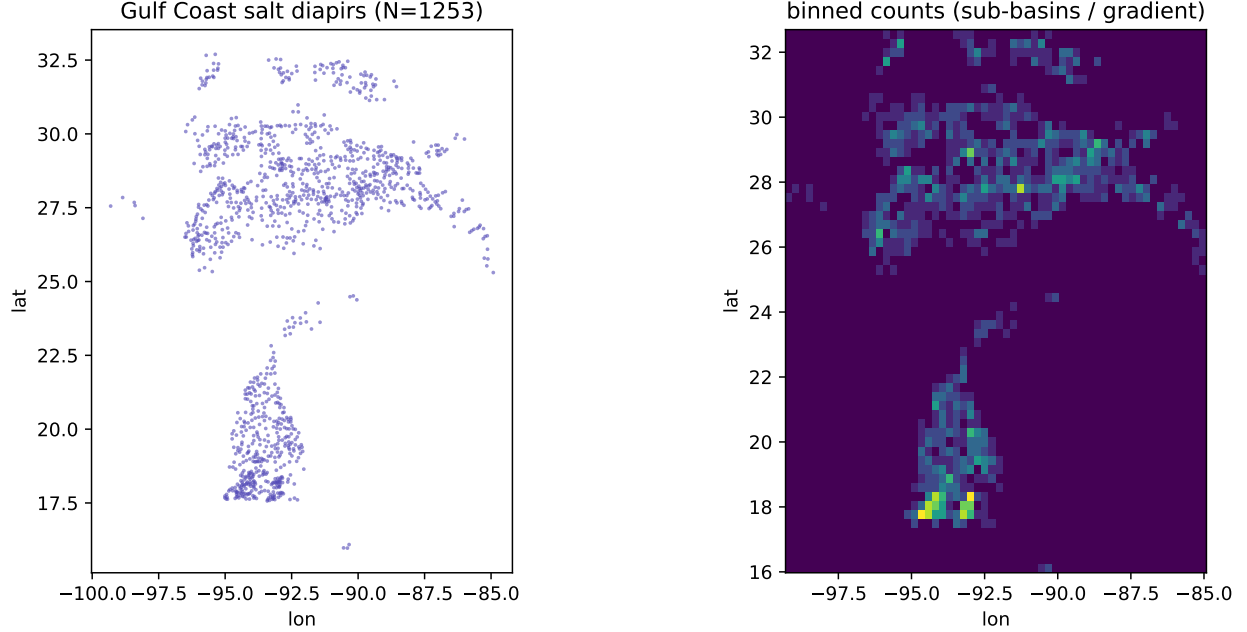
**Structure factor and number variance.** The structure factor  $S(\mathbf{k}) = N^{-1}|\sum_j e^{-i\mathbf{k}\cdot\mathbf{r}_j}|^2$  is evaluated at the wavevectors allowed by the rectangular bounding window and radially averaged; this direct estimator avoids density-field gridding artifacts. We also compute the number variance  $\sigma^2(R)$  of point counts in interior discs of radius  $R$ . Both are compared to CSR and inhomogeneous-Poisson nulls. A long-wavelength limit  $S(k \rightarrow 0) \rightarrow 0$  (equivalently  $\sigma^2(R)$  growing more slowly than the window area) would indicate hyperuniformity (Torquato, 2018); a finite  $S(k \rightarrow 0)$  with a packing peak indicates a liquid-like regular pattern.

## 4 Results

### 4.1 Gulf Coast: clustered when pooled, regular within provinces

Pooled over the whole Gulf the diapirs are strongly clustered (Clark–Evans  $R = 0.65$ ; Fig. 1), as expected for the superposition of several salt basins separated by salt-free ground. Restricting to single coherent loading-driven provinces reverses the verdict: the Texas–Louisiana shelf core ( $N = 271$ ) and the broader northern-Gulf shelf/slope and coastal province ( $N = 571$ ) are significantly *regular* ( $R = 1.12$  and  $1.10$ ; one-sided  $p = 0.002$ ; Table 1). The Bay of Campeche is indistinguishable from random, and the onshore interior salt basins, pooled across separate depocentres, are clustered (Table 1).

The pair correlation of the Texas–Louisiana province (Fig. 2) shows hard-core exclusion ( $g(r) = 0$  below  $\sim 4$  km), a clear packing peak ( $g \approx 1.5$ ) near 13 km, and damped oscillations toward unity — the classic signature of a regular/liquid-like pattern. The observed  $g(r)$  lies below both the CSR and inhomogeneous-Poisson envelopes at small  $r$ ; the global envelope test over  $r \leq 16$  km gives  $p = 0.013$  against CSR and  $p = 0.010$  against the inhomogeneous null. Correspondingly  $L(r) - r < 0$  for  $r \lesssim 13$  km. (At larger  $r$ ,  $L(r) - r$  turns positive; for this elongate shelf band this is a convex-hull window artifact rather than true clustering, and is not interpreted as long-range structure.)



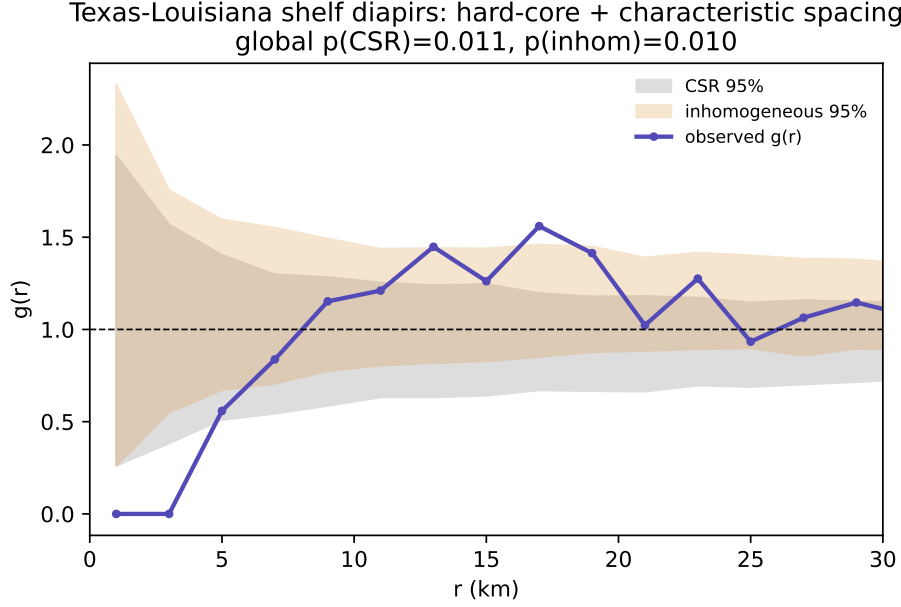
**Figure 1:** Gulf Coast salt diapirs (USGS gediapir,  $N = 1253$ ). Left: centroids of diapir polygons. Right: binned counts highlighting the separate salt basins (northern-Gulf shelf/slope and coastal trend; Bay of Campeche) and the strong basin-scale density variation. Pooled over all basins the pattern is clustered ( $R = 0.65$ ); single coherent provinces are regular (Table 1).

## 4.2 North German Basin: stronger, cleaner regularity

The North German salt structures are significantly regular already as a whole population ( $R = 1.22$ ,  $N = 693$ ;  $R = 1.21$  for the halokinetic subset). Restricting to the central basin sharpens the signal markedly: the central-basin halokinetic structures ( $N = 302$ ) give  $R = 1.41$  ( $p = 0.002$ ), and the Glückstadt Graben — the deepest, thickest-salt depocentre — gives the strongest value,  $R = 1.47$  (Table 1; Fig. 3).

The second-order statistics (Fig. 4) confirm a strong, clean regularity:  $g(r) = 0$  below  $\sim 7$  km, a packing peak near 15 km, and  $L(r) - r$  strongly negative (to  $-3.7$  km), lying outside the CSR envelope over  $r \approx 4$ –18 km before returning to  $\approx 0$  at large  $r$ . Unlike the elongate Gulf band, the compact central-basin window produces no spurious large- $r$  positivity. The global envelope test over  $r \leq 14$  km gives  $p = 0.001$  against CSR and  $p = 0.0025$  against the inhomogeneous null: the regularity is not an artifact of the basin density gradient.

A counterintuitive sub-result sharpens the interpretation. The complete halokinetic population is regular, but the mature-diapir sub-class *alone* (*Salzdiapir*,  $N = 159$ ) is *clustered* ( $R = 0.88$ ; Table 1). Pillows alone are weakly regular ( $R = 1.11$ ). Mature diapirs preferentially occupy the thickest-salt depocentres, so selecting them is a spatially biased sub-sampling that concentrates points and manufactures clustering; the regularly spaced unit is the full structure population, not a maturity class.



**Figure 2:** Pair correlation  $g(r)$  of the Texas–Louisiana shelf province with 95% Monte-Carlo envelopes for CSR (grey) and an inhomogeneous-Poisson null carrying the density gradient (orange). Hard-core exclusion below  $\sim 4$  km and a packing peak near 13 km; the observed curve falls below both envelopes at small  $r$  (global test  $p = 0.013$  vs. CSR,  $p = 0.010$  vs. inhomogeneous null).

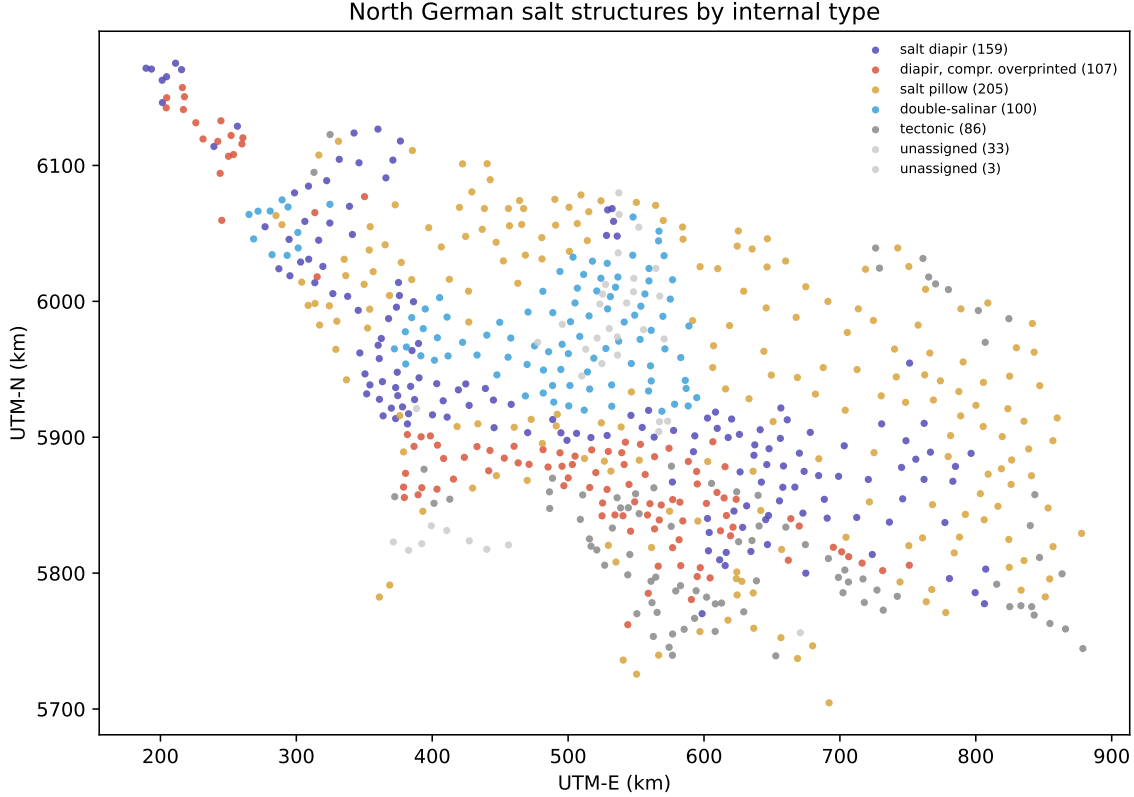
### 4.3 Character of the order: intermediate-range suppression, gradient-limited at long range

The direct structure factor of the central-basin population (Fig. 5) has three features. The single lowest accessible mode spikes to  $S \approx 3.5$ ; this is the basin-scale density gradient (salt concentrating in depocentres) and lies within the inhomogeneous-Poisson envelope. Above it,  $S(k)$  dips to a minimum  $S_{\min} \approx 0.42$  near  $k = 0.17 \text{ km}^{-1}$  (a length scale of  $\sim 37$  km, roughly three times the packing spacing), *below* the gradient-matched null — a genuine suppression of intermediate-wavelength density fluctuations. The packing peak itself is weak ( $S \approx 1.11$  at  $k = 0.55 \text{ km}^{-1}$ ). The number variance is suppressed by  $\sim 40\%$  relative to the inhomogeneous-Poisson null, but its absolute scaling is super-Poissonian ( $\sigma^2 \sim R^{2.8}$ , gradient-dominated, far from the hyperuniform  $R^1$ ) and the suppression weakens at the largest  $R$ .

The character of the salt-structure pattern is therefore a locally rigid, intermediate-range-suppressed disordered liquid: significantly more ordered than random at short and intermediate range, but *not* a confirmed hyperuniform pattern. The obstruction to a long-range verdict is physical, not statistical: the density gradient intrinsic to any salt basin (structures concentrate where salt is thickest) dominates the lowest- $k$  modes, so  $S(k \rightarrow 0) \rightarrow 0$  can be neither confirmed nor excluded from map data alone.

## 5 Discussion

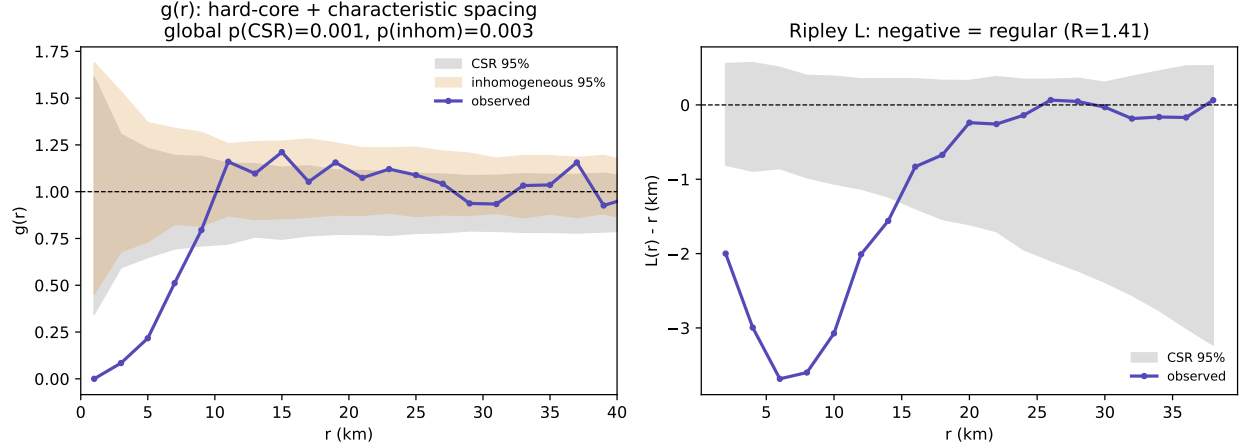
**A map-view Rayleigh–Taylor signature.** Within coherent loading-driven provinces, salt-structure populations in both basins are significantly regularly spaced, with hard-core exclusion and a 12–15 km characteristic spacing. This is the spatial expression expected of RT wavelength



**Figure 3:** North German salt structures (BGR InSpEE, 693 structures) coloured by internal type. Diapirs, compressively overprinted diapirs, pillows and double-salinar structures constitute the halokinetic population; double-salinar structures and many diapirs concentrate in the deep central basin / Glückstadt Graben, where regularity is strongest.

selection combined with salt-withdrawal exclusion. The hard-core component has a concrete halokinetic origin: as a structure grows it withdraws salt from its surroundings, thinning the mother-salt layer into a peripheral rim syncline that may weld out entirely as a primary salt weld (Hudec & Jackson, 2007; Jackson & Hudec, 2017). Where the source layer is locally exhausted a neighbouring structure cannot nucleate, so the withdrawal aureole acts as an exclusion radius — the map-view counterpart of the observed  $g(r) = 0$  at small separations — while the finite RT wavelength sets the dominant spacing between structures. Two observations argue for an RT origin rather than purely structural control. First, the result replicates across two independent basins and tectonic settings. Second, regularity strengthens systematically with salt thickness — weakest on the Gulf shelf, stronger in the central North German basin, strongest in the deep, thick-salt Glückstadt Graben — consistent with the RT prediction that the selected wavelength (and hence the clarity of a single dominant spacing) increases with salt-layer thickness. Purely fault-localised ascent, by contrast, would align and cluster structures, as is indeed seen where structures are pooled across sub-basins or where ascent follows inherited faults.

**Trigger versus spacing control.** This is not a claim that salt ascent is independent of structural control. Extension and basement faulting commonly trigger or localise diapirism — reactive diapirism, in which a fault provides the initial weakness for salt to exploit (Vendeville & Jackson, 1992) — and where ascent remains dominated by inherited faults the structures align and cluster

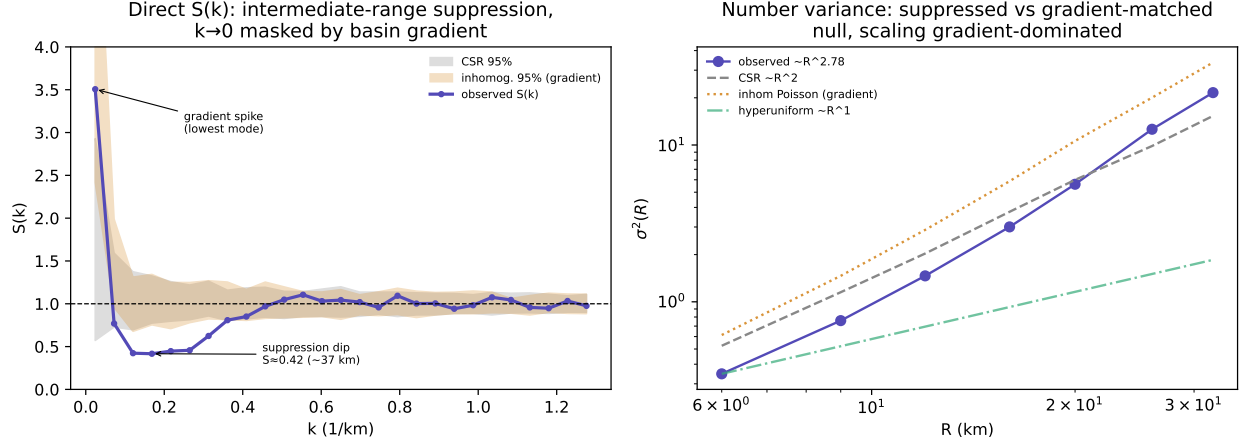


**Figure 4:** Central-basin halokinetic structures ( $N = 302$ ). Left:  $g(r)$  with CSR and inhomogeneous-Poisson 95% envelopes (hard-core to  $\sim 7$  km, packing peak near 15 km; global test  $p = 0.001$  vs. CSR,  $p = 0.0025$  vs. inhomogeneous null). Right:  $L(r) - r$  strongly negative (regular) over  $r \approx 4$ –18 km, returning to  $\approx 0$  at large  $r$ .

rather than disperse. Our own data show exactly this: the pooled onshore interior basins are clustered ( $R = 0.82$ ), and salt structures in the fault-influenced parts of the basins do not contribute the regular signal. The claim here is therefore narrower and explicitly setting-dependent: *within loading-dominated provinces*, whatever the initial trigger, the characteristic spacing of the mature structure population is governed by the buoyant (RT) wavelength and salt-withdrawal exclusion, yielding a statistically regular map pattern; faulting may set where structures begin, but buoyant wavelength selection sets how far apart they end up. The regularity is thus an emergent property of the loading-driven provinces, not a universal property of salt tectonics.

**Three methodological cautions.** The analysis exposes three ways to reach a wrong verdict from the same data. (i) *Superposition*: pooling distinct sub-basins superposes independent processes and manufactures apparent clustering (the whole-Gulf  $R = 0.65$ , the pooled onshore interior  $R = 0.82$ ); inference must be made within single provinces. (ii) *Sub-class selection*: analysing only mature diapirs returns clustering ( $R = 0.88$ ) because maturity is spatially biased toward depocentres; the regularly spaced unit is the complete structure population, not a maturity class. (iii) *Density gradient*: the basin-scale concentration of salt in depocentres injects low-wavenumber power that masks the long-wavelength behaviour and biases the statistics toward clustering. We control for the first by province subsetting, the second by analysing the full halokinetic population, and the third by testing against an inhomogeneous-Poisson null — against which the detected regularity, being conservative, survives.

**Character and limits.** The two families of statistic deliberately probe different scales: the Clark–Evans index and the small- $r$  pair correlation measure short-range, hard-core exclusion, whereas the low-wavenumber limit of  $S(k)$  and the large- $R$  number variance measure the long-range suppression of density fluctuations. The salt-structure populations are unambiguous on the former — they are significantly short-range regular — and gradient-limited on the latter. The structure factor shows that the order extends beyond nearest neighbours: density fluctuations are suppressed at intermediate wavelengths ( $\sim 3$  times the packing spacing) below the gradient-matched null. But the long-wavelength limit, and hence any claim of hyperuniformity, is inaccessible from map data



**Figure 5:** Left: direct structure factor  $S(k)$  of the central-basin population with CSR and inhomogeneous-Poisson 95% envelopes. A gradient spike at the lowest mode, a suppression dip ( $S \approx 0.42$  at  $\sim 37$  km scale) below the gradient-matched null, and a weak packing peak. Right: number variance  $\sigma^2(R)$ ; suppressed relative to the inhomogeneous-Poisson null but with gradient-dominated  $\sim R^{2.8}$  scaling, not the hyperuniform  $R^1$ .

because the intrinsic basin density gradient dominates the lowest- $k$  modes. The salt-structure populations are thus best described as locally rigid, intermediate-range-suppressed disordered liquids. The same point-pattern and structure-factor diagnostics used here are those applied to disordered, fluctuation-suppressed systems generally (Torquato, 2018); their application to salt basins is, to our knowledge, new, and provides a reproducible, statistically hardened way to quantify halokinetic self-organisation in map view.

## 6 Conclusions

- Salt-structure populations in coherent loading-driven provinces of the U.S. Gulf Coast and the North German Basin are significantly regularly spaced (Clark–Evans  $R = 1.10$ – $1.47$ ,  $p \approx 0.002$ ), with hard-core exclusion and a 12–15 km characteristic spacing, robust to a density-gradient inhomogeneous-Poisson null.
- Regularity strengthens with salt thickness (Glückstadt Graben > central basin > Gulf shelf), consistent with Rayleigh–Taylor wavelength selection plus salt-withdrawal exclusion.
- The order is real at short and intermediate range, but long-range hyperuniformity cannot be confirmed: the intrinsic basin density gradient masks the long-wavelength limit.
- Whole-basin pooling and mature-diapir-only sub-sampling both spuriously return clustering; correct inference requires single-province analysis of the complete structure population.

## Data availability

U.S. Gulf Coast salt diapirs: USGS data release *Salt Diapirs in the Gulf Coast* (gcdiapirog), doi:10.5066/P9KH3WB4 (public domain). North German salt structures: BGR *InSpEE-Salzstrukturen* (© BGR, Hannover, 2015), open geodata licence (geonutzv); base mapping after Reinhold et al. (2008). Analysis used only open scientific Python (NumPy, SciPy, Matplotlib). All



**Table 1:** Clark–Evans index  $R$  (edge-corrected against a matched CSR Monte-Carlo null,  $S = 499$ ) and one-sided dispersion  $p$ .  $R > 1$ : regular/dispersed;  $R \approx 1$ : random;  $R < 1$ : clustered. NND = median nearest-neighbour distance.

| Province / subset                        | $N$  | NND (km) | $R$  | verdict                 |
|------------------------------------------|------|----------|------|-------------------------|
| <i>U.S. Gulf Coast (USGS gcdiapirog)</i> |      |          |      |                         |
| Whole basin (all pooled)                 | 1253 | 11.1     | 0.65 | clustered               |
| N. Gulf shelf/slope + coast              | 571  | 11.6     | 1.10 | regular ( $p = 0.002$ ) |
| Texas–Louisiana shelf core               | 271  | 12.0     | 1.12 | regular ( $p = 0.002$ ) |
| Bay of Campeche                          | 366  | 9.2      | 1.02 | random                  |
| Onshore interior basins (pooled)         | 146  | 13.5     | 0.82 | clustered               |
| <i>North German Basin (BGR InSpEE)</i>   |      |          |      |                         |
| All structures                           | 693  | 10.1     | 1.22 | regular ( $p = 0.002$ ) |
| All halokinetic                          | 571  | 11.1     | 1.21 | regular ( $p = 0.002$ ) |
| Central basin, halokinetic               | 302  | 10.3     | 1.41 | regular ( $p = 0.002$ ) |
| Glückstadt Graben, halokinetic           | 109  | 11.0     | 1.47 | regular ( $p = 0.002$ ) |
| Salt diapirs only (mature)               | 159  | 11.1     | 0.88 | clustered               |
| Salt pillows only                        | 205  | 15.0     | 1.11 | regular ( $p = 0.004$ ) |

analysis code, the derived coordinate datasets, the figure-generation scripts and the numeric results are openly available at <https://github.com/Ruqing1963/salt-structure-spacing>.

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